

Global Valuation — Data Exploration

Schema, Coverage, and Preliminary Diagnostics

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Data Overview

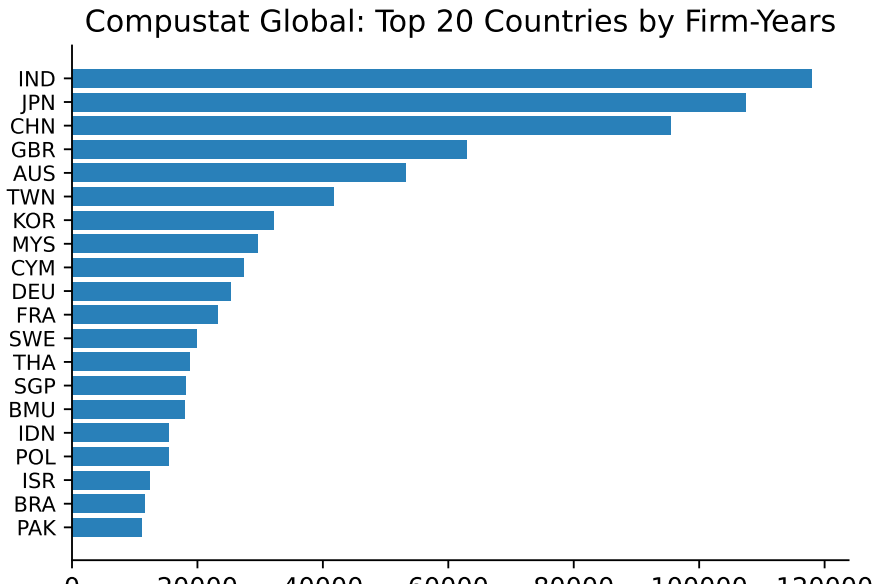
Datasets

File	Size	Rows	Cols	Description
compustat_global	2.3 GB	981K	468	Global annual fundamentals (124 countries)
compustat_america	0.8 GB	254K	984	NA fundamentals + CRSP link
returns_global	1.9 GB	10.4M	25	Global monthly prices
returns_america	0.9 GB	3.6M	55	CRSP US monthly returns

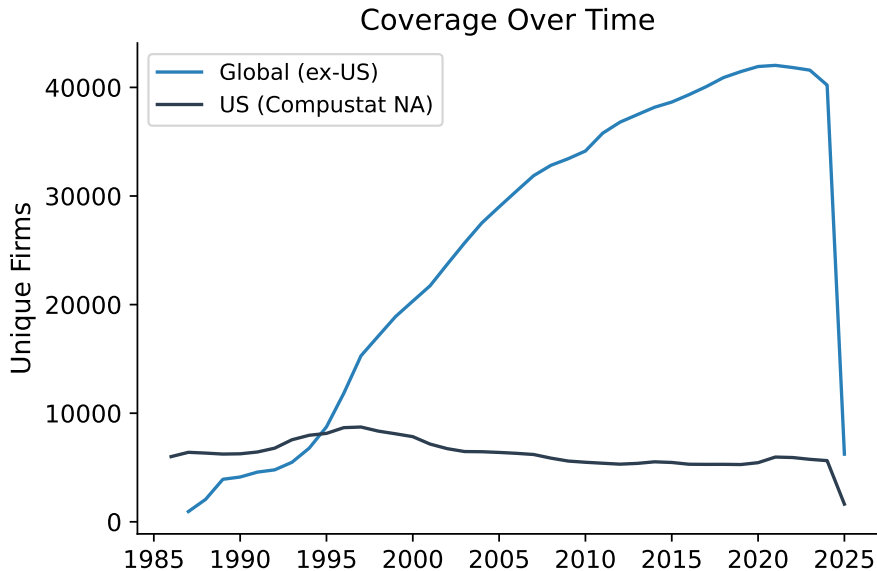
- Compustat Global: fiscal years 1987–2025
- Compustat America: fiscal years 1986–2025
- Returns America: 1986–2024, Returns Global: 2007–2026

Coverage

Top 20 Countries by Firm-Years (Compustat Global)



Coverage Over Time



Key Variables

Missingness (% NaN)

Compustat Global

Variable	% Missing
at (assets)	0.3%
ceq (equity)	0.4%
sale	18.3%
ebitda	1.2%
mkvalt	not in file

Compustat America

Variable	% Missing
at (assets)	8.4%
ceq (equity)	8.5%
sale	8.8%
ebitda	11.4%
prcc_f	0.4%
csho	0.6%

Note: mkvalt absent from Global file — market cap must be computed by merging with returns_global (price $\times$ shares).

Data Quality

Data Quality Flags

Compustat Global

Flag	Count
Zero total assets	7,242
Negative total assets	4
Negative equity	33,321

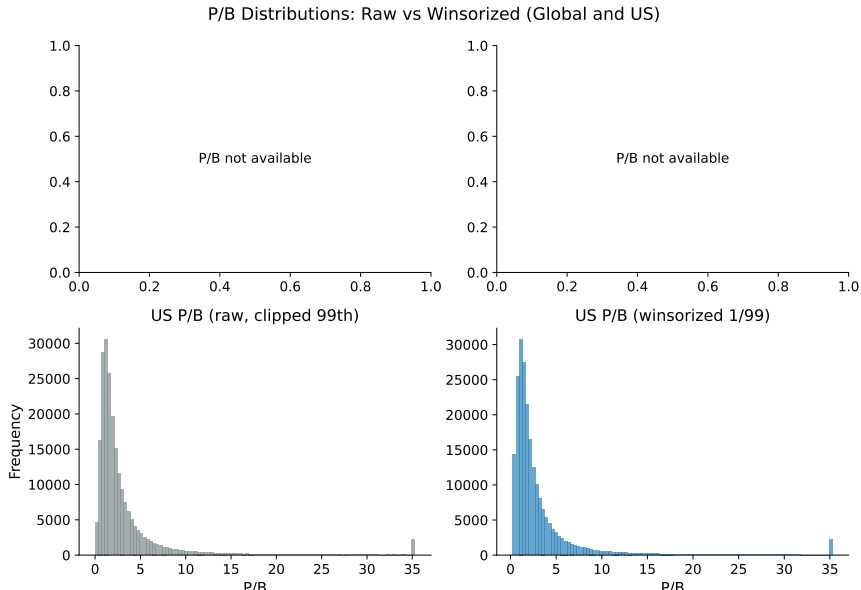
Compustat America

Flag	Count
Zero total assets	20
Negative total assets	0
Negative equity	10,339

- Negative equity: 3.4% of Global, 4.1% of US — must filter for P/B
- Zero-asset firms likely shell companies or data errors

Valuation Distributions

P/B Distributions: Raw vs Winsorized



Returns

Returns Summary

Returns America (CRSP)

Statistic	Value
Unique securities	33,074
Date range	1986–2024
Non-missing RET	98.5%
Mean monthly	0.91%
Median monthly	0.19%
Std dev	18.3%

Returns Global

Statistic	Value
Unique securities	58,261
Date range	2007–2026
Return column	none
Price column	prccm

Returns must be computed from monthly prices.

Key Takeaways

Pitfalls and Next Steps

Pitfalls identified:

1. `gvkey` is lowercase in Global, **uppercase** (`GVKEY`) in America
2. `mkvalt` only in America — Global needs `price × shares` merge
3. Returns Global has **no return column** — compute from `prccm`
4. 43K negative-equity observations to handle for P/B ratios
5. Max US monthly return = 3,900% — outlier screening needed

Next steps:

1. Build cleaning pipeline (filter, winsorize, harmonize identifiers)
2. Merge Compustat with returns for market-cap-based ratios
3. Compute country-level valuation multiples (P/B, EV/EBITDA)